

Title

Protocol for an exploratory analysis of hypertension-related treatment heterogeneity in anticoagulation for cryptogenic stroke with atrial cardiopathy

Abstract

Background: Randomized trials of anticoagulation versus antiplatelet agents in cryptogenic stroke have yielded conflicting results, possibly due to misclassification of stroke aetiology. Hypertension-related small vessel disease may differ mechanistically from cardioembolic mechanisms. **Objective:** We describe a protocol for exploratory analysis of a completed randomized trial examining whether severity of hypertension modifies treatment effect of anticoagulation versus antiplatelet therapy. **Methods:** Analysis will include 945 participants from the parent randomized trial (from 185 North American sites, 2018-2023) with recent cryptogenic stroke, documented atrial cardiopathy on imaging, and available baseline hypertension data. Hypertension with high-risk features is defined as: systolic blood pressure ≥ 160 mm Hg at randomization, left ventricular hypertrophy on echocardiography, or both. **Primary outcome:** recurrent ischemic stroke or systemic embolism. Cox proportional hazards models will examine treatment-by-hypertension interaction. **Hypothesis:** anticoagulation (apixaban) provides benefit in patients without advanced hypertension but not in those with hypertension-related high-risk features.